From Axiomatic Necessity (F_{min}) to Measurable Throughput (Ψ)

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1. Axiomatic Necessity and the Non-Derivability of Closure

The Theorem of Axiomatic Necessity (TNA) establishes a general result:

> *Any system that maintains internal coherence over time requires at least one external, non-derivable foundational constraint (denoted F_{min}).*

This statement is **metamathematical**, not physical. It does not specify the nature of F_{min}; it only asserts that purely internal rules are insufficient to guarantee sustained coherence.

Crucially, the problem addressed by TNA is **temporal**. The failure mode is not logical inconsistency but **dynamic degradation**: without F_{min}, coherence decays as the system evolves.

Therefore, F_{min} cannot be interpreted as a static axiom or fixed truth-value. A static element cannot counteract time-dependent entropy accumulation.

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2. From Static Foundation to Operational Constraint

To fulfill its role, F_{min} must be instantiated as an **operation** acting on the system state.

We denote this operational realization as N_1:

> *N_1 is a non-derivable operator whose function is to maintain structural coherence by counteracting entropy accumulation.*

No semantic or intentional content is assumed. N_1 is defined **purely operationally**: it is whatever process prevents uncontrolled growth of internal incoherence.

At this stage, the framework remains within **systems theory and information dynamics**.

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3. Physical Constraint: Any Real Operation Has Thermodynamic Cost

A well-established result in physics (e.g., Landauer's principle and non-equilibrium thermodynamics) is that:

> *Any physical process that performs information selection, state reduction, or structural maintenance incurs a thermodynamic cost.*

In particular:

- maintaining correlations,
- suppressing disorder,
- enforcing constraints on accessible states,

all require **entropy export** to the environment.

Thus, if N_1 is instantiated in any physical system (biological, cognitive, social, or engineered), it must be accompanied by a measurable entropy flow.

This is not an additional assumption; it is a physical constraint.

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4. Definition of Throughput Ψ

We define the system's operational negentropy throughput Ψ as:

> *The maximum rate at which a system can export entropy in order to preserve internal coherence.*

Formally, entropy S has units of J/K. Its time derivative,

$$\dot{S} = \frac{dS}{dt},$$

has units:

$$\frac{\mathrm{J}}{\mathrm{K \cdot s}} = \frac{\mathrm{W}}{\mathrm{K}}.$$

Therefore, Ψ is naturally expressed in **W/K**.

No reinterpretation of physical entropy is required; this is standard thermodynamic bookkeeping.

5. Incoherence Functional and Stability Condition

Let $G(t)$ denote the rate at which incoherence (unprocessed gradient, disturbance, or constraint load) enters the system.

Let $\aleph$ represent accumulated incoherence.

Then:

$$\frac{d\aleph}{dt} = G(t) - \Psi(t)$$

- If $\Psi(t) \geq G(t)$, incoherence does not accumulate.
- If $\Psi(t) < G(t)$ over extended periods, $\aleph$ grows and structural failure becomes unavoidable.

This formulation is agnostic to system domain and applies equally to:

- thermodynamic systems,
- biological metabolism,
- neural processing,
- urban infrastructure,
- institutional governance.

The variables change interpretation, not structure.

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6. Clarification on "Creatio Continua"

The framework does **not** posit continuous creation of energy or information.

"Creatio continua" is used in a strictly operational sense:

> *the continuous expenditure of work required to prevent structural decay in an open system.*

What is measured is **not creation**, but the **cost of persistence**.

The quantity measured in W/K is the entropy export rate required to maintain a given level of organization.

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7. Summary (Minimal Claim Set)

1. Sustained coherence requires a non-derivable constraint ($F_{\min}$).
2. In time-evolving systems, this constraint must be operational (N_1).
3. Any real operation enforcing coherence has a thermodynamic cost.
4. That cost is an entropy export rate.
5. The corresponding throughput Ψ is naturally measured in W/K.

No metaphysical assumptions are introduced. The argument relies only on:

- metamathematical necessity,
- operational definitions,
- established physical constraints.

Suggested neutral closing sentence (safe for reviewers)

> *Under minimal assumptions, the axiomatic requirement for external coherence implies an operational constraint whose physical instantiation necessarily entails a measurable entropy throughput. The parameter Ψ, expressed in W/K, quantifies the thermodynamic cost of maintaining structure in time.*

Appendix A

Thermodynamic Grounding of Throughput (Ψ): Connections to Landauer, Prigogine, and Non-Equilibrium Steady States

A.1. Information Processing and Landauer's Bound

Landauer's principle establishes a lower bound on the thermodynamic cost of information erasure:

$$\Delta Q \geq k_B T \ln 2$$

per bit of logically irreversible operation, where ΔQ is dissipated heat and T the ambient temperature.

This result implies a general constraint:

> *Any physical process that reduces the space of accessible system states must dissipate entropy to the environment.*

In the context of TNA, the operator N_1 performs precisely such a function: it restricts state proliferation by enforcing coherence conditions that are not derivable from internal dynamics alone.

Importantly, the framework does **not** assume digital computation. Landauer's bound applies to *any* logically irreversible constraint, including:

- selection among competing trajectories,
- suppression of fluctuations,
- maintenance of correlations.

Therefore, the continuous operation of N_1 entails a **minimum entropy export rate**, regardless of implementation details.

A.2. Open Systems and Prigogine's Dissipative Structures

Prigogine's theory of dissipative structures shows that:

> *Ordered macroscopic structures can persist only in systems driven far from equilibrium and coupled to entropy sinks.*

Formally, for an open system:

$$\frac{dS_{sys}}{dt} = \sigma - \Phi$$

where:

- $\sigma \geq 0$ is internal entropy production,
- Φ is entropy flow to the environment.

Structural persistence requires:

$$\Phi \geq \sigma$$

In TNA notation:

- accumulated incoherence corresponds to $\aleph$,
- internal entropy production contributes to $G(t)$,
- entropy export capacity corresponds to $\Psi(t)$.

Thus, the TNA stability condition:

$$\frac{d\aleph}{dt} = G(t) - \Psi(t) \leq 0$$

is a direct structural analog of Prigogine's balance equation.

No new thermodynamic principle is introduced; the framework recasts known results in throughput form.

A.3. Non-Equilibrium Steady States (NESS)

A non-equilibrium steady state is defined by:

$$\frac{dS_{sys}}{dt} = 0 \quad \text{with} \quad \sigma > 0$$

Such states are maintained by continuous entropy export.

Key properties of NESS relevant to TNA:

1. **Time asymmetry**: steady state does not imply reversibility.
2. **Sustained dissipation**: order exists only while throughput persists.
3. **Threshold behavior**: below a minimum flux, the state collapses.

This directly motivates the definition of Ψ as a **capacity**, not merely a flux. Systems fail not when entropy production increases, but when **export capacity saturates**.

Collapse corresponds to loss of NESS, not equilibrium attainment.

A.4. Units and Operational Meaning of Ψ

Entropy has units of J/K. Entropy flow rate therefore has units:

$$[\Psi] = \frac{\mathrm{J}}{\mathrm{K{\cdot}s}} = \frac{\mathrm{W}}{\mathrm{K}}$$

This unit choice is not symbolic. It reflects:

- power expenditure per unit thermodynamic slack,
- capacity to process gradients without accumulation.

Ψ is not "negative entropy" in an ontological sense; it is **operational negentropy**, measurable as sustained entropy export per unit temperature.

A.5. Relation to the Axiomatic Foundation $F_{\min}$

The metamathematical result of TNA states that coherence cannot be self-generated.

The thermodynamic interpretation is conservative:

> *If coherence is not derivable internally, it must be enforced by a process that reduces accessible state space, and such enforcement necessarily dissipates entropy.*

Thus:

- $F_{\min}$ is abstract and non-physical,

- N_1 is its operational instantiation,
- Ψ is the measurable thermodynamic footprint of that instantiation.

No claim is made that axioms themselves "emit watts." Only that **their physical enforcement does**.

A.6. Scope and Limits

This appendix does **not** claim:

- equivalence between logical axioms and physical forces,
- reduction of cognition or institutions to thermodynamics,
- universality beyond open, time-evolving systems.

It claims only that:

> *Any real system that maintains structure against entropic drift must satisfy the same throughput constraints as dissipative physical systems.*

A.7. Concluding Statement

The connection between TNA and thermodynamics does not rest on metaphor but on shared necessity conditions. Landauer provides the lower bound, Prigogine the structural context, and NESS the dynamical regime. The parameter Ψ unifies these results as an operational measure of coherence maintenance.